LAXMI CHARITABLE TRUST'S

SHETH L.U.J. COLLEGE OF ARTS & SIR M.V. COLLEGE OF SCIENCE AND COMMERCE

DEPARTMENT OF INFORMATION TECHNOLOGY

AI MINI PROJECT DOCUMENTATION

IRIS FLOWER SPECIES CLASSIFICATION USING MACHINE LEARNING

Student Name	Ritika Rai / Anurag Yadav
Roll No.	T035 / TO51
Class / Batch	T.Y BSc-IT
Subject	Artificial Intelligence
Academic Year	2026-2027

1. Introduction

The project “Iris Flower Species Classification Using Machine Learning” is an Artificial Intelligence and Machine Learning project developed using Python and Scikit-learn. The system predicts the species of an Iris flower based on four measurements: sepal length, sepal width, petal length and petal width. Machine Learning is used to learn patterns from previously available flower data and classify a new flower into one of three species: Iris Setosa, Iris Versicolor or Iris Virginica. The project demonstrates the basic AI workflow of data collection, preprocessing, training, testing, prediction and performance evaluation.

2. Problem Statement

Identifying the species of an Iris flower manually based on its physical measurements can require knowledge of botanical characteristics. The aim of this project is to develop a Machine Learning model that can automatically classify an Iris flower into its correct species based on its sepal and petal measurements.

3. Objectives

- To collect and understand the Iris flower dataset.
- To preprocess and prepare the dataset for Machine Learning.
- To train a Machine Learning classification model.
- To test the trained model using unseen data.
- To predict the species of new Iris flowers and evaluate model performance.

4. Technologies Used

Technology	**Purpose**
Python	Programming language
Jupyter Notebook / Google Colab	Development environment
Pandas	Data handling and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Graphical visualization
Scikit-learn	Machine Learning
Joblib	Saving trained model

5. Dataset

Dataset Name: Iris Flower Dataset

Source: The Iris dataset is a standard Machine Learning dataset and is available directly through Scikit-learn.

Number of Records: 150

Number of Features: 4

Important Attributes / Features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable: Species

Target classes:

- Iris Setosa
- Iris Versicolor
- Iris Virginica

6. Methodology / Working

Dataset Collection: The Iris dataset is loaded using Scikit-learn.

Data Understanding: The dataset is examined to understand its features, target classes and statistical information.

Data Preprocessing: The data is checked and prepared for Machine Learning.

Train-Test Split: The dataset is divided into 80% training data and 20% testing data.

Model Training: A Decision Tree Classifier is trained using the training data.

Prediction: The trained model predicts the species of flowers from the testing dataset.

Evaluation: Predictions are evaluated using accuracy, precision, recall, F1-score and confusion matrix.

New Prediction: The trained model can predict the species of a new Iris flower using its four measurements.

7. Algorithm Used

Decision Tree Classifier

A Decision Tree is a supervised Machine Learning algorithm used for classification and regression problems. In this project, the Decision Tree learns decision rules from the four flower measurements and uses these rules to classify flowers into their respective species.

Why Decision Tree is Used:

- It is easy to understand.

- It works well for classification problems.
- Its decision-making process can be visualized.
- It can handle numerical features effectively.
- It is suitable for a beginner-level AI project.

How It Works:

The algorithm repeatedly divides the dataset based on feature values. It selects useful feature-based conditions and follows branches until reaching a leaf node that represents the predicted flower species.

8. Implementation

The following Python implementation covers dataset loading, train-test splitting, model training, prediction and evaluation.

8.1 Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

8.2 Load Dataset

```
iris = load_iris()

X = iris.data
y = iris.target

print("Dataset loaded successfully")
print("Number of records:", len(X))
```

8.3 Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42
)
```

8.4 Create and Train Model

```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

8.5 Make Predictions

```
y_pred = model.predict(X_test)
```

8.6 Calculate Accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

8.7 Classification Report

```
print(classification_report(y_test, y_pred))
```

8.8 Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, cmap="Blues", fmt="d")

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```

8.9 Predict a New Flower

```
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("Predicted Species:",
      iris.target_names[prediction][0])
```

9. Results

After training and testing the model, the system produces a prediction for the species of an Iris flower. The exact accuracy should be recorded from the student's own execution because it can vary with the train-test split and model configuration.

- **Dataset output:**

```
========== DATASET OUTPUT ==========

First 10 records:
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm Species
0            5.1           3.5            1.4           0.2  setosa
1            4.9           3.0            1.4           0.2  setosa
2            4.7           3.2            1.3           0.2  setosa
3            4.6           3.1            1.5           0.2  setosa
4            5.0           3.6            1.4           0.2  setosa
5            5.4           3.9            1.7           0.4  setosa
6            4.6           3.4            1.4           0.3  setosa
7            5.0           3.4            1.5           0.2  setosa
8            4.4           2.9            1.4           0.2  setosa
9            4.9           3.1            1.5           0.1  setosa
```

- **Train-test split:**

```
========== TRAIN-TEST SPLIT ==========
Total records: 150
Training records: 120
Testing records: 30
Training percentage: 80.0 %
Testing percentage: 20.0 %
```

- **Model training:**

```
========== MODEL TRAINING ==========
Algorithm: Decision Tree Classifier
Training status: Model trained successfully
Number of training samples: 120
Number of features: 4
```

- **Accuracy output:**

```
========== ACCURACY ==========
Accuracy: 1.0
Accuracy Percentage: 100.0 %
```

- **Classification report:**

```
========== CLASSIFICATION REPORT ==========
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

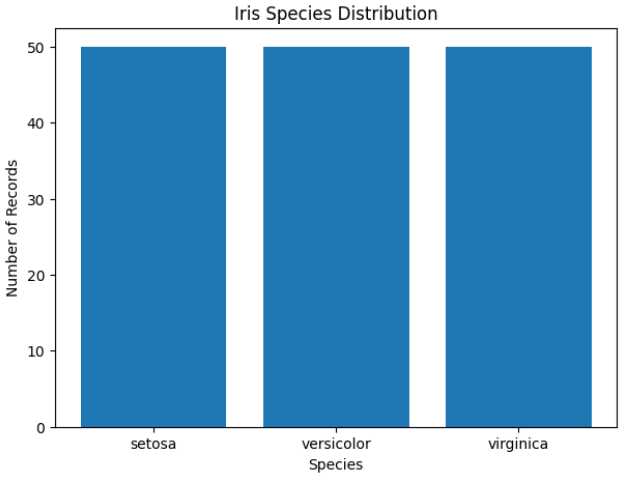
- **Confusion matrix:**

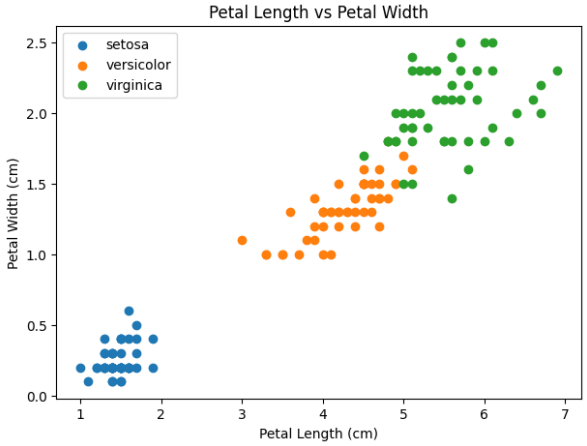
```
========== CONFUSION MATRIX ==========
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

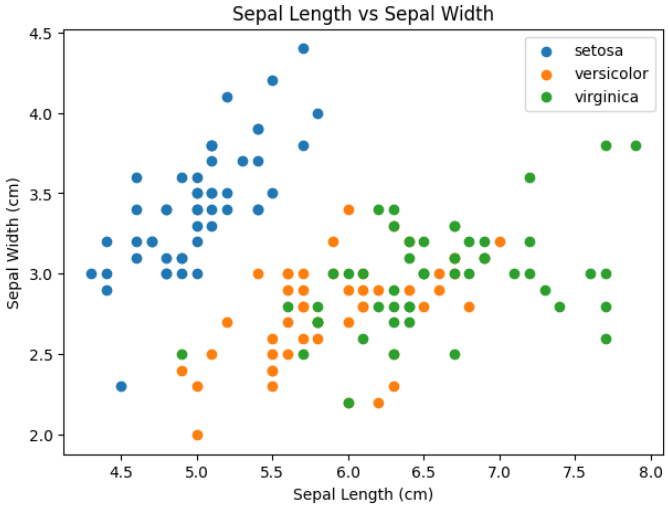
- **Sample prediction:**

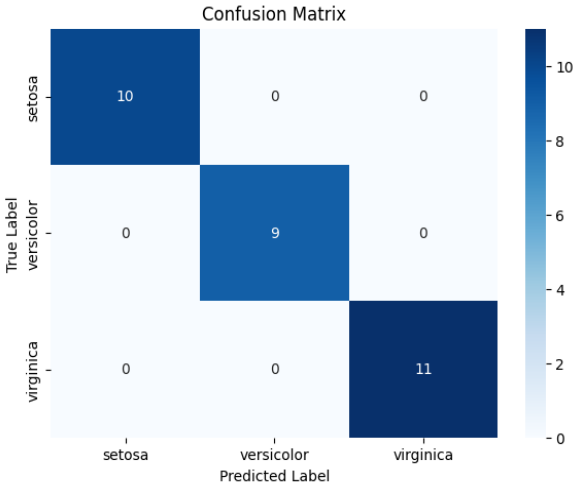
```
========== SAMPLE PREDICTION ==========
Input values:
Sepal Length = 5.1 cm
Sepal Width  = 3.5 cm
Petal Length = 1.4 cm
Petal Width  = 0.2 cm
```

- **Graphs/visualizations:**









10. Advantages

- Simple and easy to use – the system can classify flowers automatically.
- Fast prediction – once trained, the model can quickly predict new samples.
- Reduces manual classification – the system automatically identifies the species.
- Easy to understand – Decision Tree provides an interpretable classification process.
- Educational – demonstrates the complete Machine Learning workflow.

11. Limitations

- The Iris dataset is relatively small compared with real-world datasets.
- The system only recognizes three Iris species.
- The model depends on the quality of the training data.
- It may not perform well on flower species that are not represented in the dataset.

12. Applications

- Botanical research
- Educational Machine Learning systems
- Plant identification applications
- Agriculture-related classification systems
- AI and Data Science learning platforms
- Automated species classification research

13. Conclusion

The Iris Flower Species Classification Using Machine Learning project successfully demonstrates how Artificial Intelligence can be used to classify data automatically. A Decision Tree Machine Learning model was trained using Iris flower measurements such as sepal length, sepal width, petal length and petal width. The trained model was then used to predict the species of new flowers. The project provides practical understanding of dataset preparation, preprocessing, model training, prediction and performance evaluation, making it suitable as a TY B.Sc.IT Artificial Intelligence mini project.

14. Future Scope

- Adding more flower species and larger datasets.
- Comparing different Machine Learning algorithms such as KNN, Random Forest, Logistic Regression and SVM.
- Developing a graphical user interface.
- Creating a web application for real-time flower prediction.
- Deploying the trained AI model online.
- Adding image-based flower classification using Deep Learning.

15. References

1. **Scikit-learn Documentation – Iris Dataset**
   Used for the Iris dataset and Machine Learning implementation.

2. **Scikit-learn Documentation – Decision Tree Classifier**
   Used as a reference for implementing the Decision Tree algorithm.

3. **GitHub – Iris Classification using Scikit-learn**
   Used as a reference for the project structure, classification workflow and evaluation.